Bean Leaf Disease Classification with SigLIP 2 and DINOv2: Accuracy, Adaptation and Cost

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MAAI901 Deep Learning for Visual Data

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1. Introduction and research question

This study compares SigLIP 2, a recent vision-language encoder (2025), with DINOv2 (2023) to classify bean leaves [1,2]. Both models use attention over image-patch representations learned during pretraining.

The primary objective was to determine whether partially fine-tuned SigLIP 2 achieves higher macro-F1 than partially fine-tuned DINOv2. Secondary objectives concerned frozen features, adaptation benefits, computational cost and failures. This was exploratory because the original DINOv2 test results were already known. Each configuration started from its model's original pretrained encoder weights under fixed settings. Reusing the test data means these results cannot provide fresh confirmatory evidence.

2. Dataset selection and audit

The Beans dataset from Makerere AI Lab contains smartphone photographs collected from Ugandan fields with the National Crops Resources Research Institute. Agricultural experts labelled them as angular leaf spot, bean rust or healthy [3]. The small dataset made pretrained models useful. Field backgrounds and varied lighting also provide a more realistic classification challenge than isolated leaf cutouts.

The pinned Hugging Face release contains 1,295 images, although the original repository mentions 1,296 [3,4]. Table 1 shows the loaded counts. All images were 500 × 500 RGB, with none corrupt. File and decoded-pixel hashes found no exact duplicates. A 64-bit perceptual hash with Hamming distance up to 12 identified 14 candidate pairs, including seven across splits. AI-assisted visual checking judged them different, so none were removed. This was not independent expert verification of labels.

Class	Training	Validation	Test
Angular leaf spot	345	44	43
Bean rust	348	45	43
Healthy	341	44	42
Total	1,034	133	128

Table 1. Image counts in the audited dataset release.

The published split was retained. Test images had also been inspected during the original integrity check. Without plant or farm IDs, independence at those levels could not be verified. Hash checks cannot rule out all related images. Overlap with either model's pretraining data was not measured. Dataset revisions, the manifest and screening decisions accompany the code.

3. Model selection and configuration

Google's public `google/siglip2-base-patch16-224` checkpoint was downloaded without authentication, and its published weight hash was verified [5]. SigLIP 2 combines image-text learning, captioning, self-distillation, masked prediction and data curation [1]. Its paper reports 78.2% ImageNet zero-shot accuracy versus 76.2% for original SigLIP, leading the same-size B/16-224 baselines in the paper's Table 1 [1]. This supports choosing a recent advanced model, but does not establish the best October 2026 model or state-of-the-art Beans performance.

Only SigLIP 2's image tower was used. Its 12 blocks process 196 patches of 16 × 16 pixels at 224 × 224 resolution, with 768-dimensional embeddings and 12 attention heads. Final normalisation precedes pretrained attention pooling, producing a vector for a new three-class linear classifier [5,6]. Token zero is an image patch, not a class token. This fixed-resolution checkpoint uses `SiglipVisionModel`; NaFlex was not used.

The `facebook/dinov2-small` model has 12 blocks, 384-dimensional embeddings, six attention heads and 256 patches of 14 × 14 pixels, plus a class token. Its final normalised class token feeds the classifier [2,7]. DINOv2 was pretrained on LVD-142M; SigLIP 2 used WebLI. Differences in capacity, pretraining, pooling and preprocessing mean this comparison cannot isolate publication year or architecture.

Model	Total parameters	Frozen-arm trainable	Partial-arm trainable
DINOv2-S/14	22,057,731	1,155	3,552,387
SigLIP 2-B/16	92,886,531	2,307	14,179,587

Table 2. Loaded image-model parameters, including pooling and the new classifier. The text tower is excluded.

4. Pre-processing and training

RGB images were resized in full to 224 × 224. SigLIP 2 used bilinear interpolation and channel mean/standard deviation 0.5 [5]. DINOv2 used bicubic interpolation, means (0.485, 0.456, 0.406) and standard deviations (0.229, 0.224, 0.225). Its official resize-to-256 then centre-crop-to-224 procedure was replaced with full-image resizing to retain outer regions [7]. Both approaches lose fine detail.

Training used horizontal and vertical flips, each with probability 0.5, followed by uniform rotation within ±15° and model-specific mean-colour fill. This varied orientation without deliberately changing lesion colour or cropping symptoms. Seed-, epoch- and image-dependent randomness matched sample order and augmentation decisions across configurations. Validation/testing used deterministic preprocessing. Frozen features were calculated from augmented training images.

The frozen setting trained only the new classifier. Partial fine-tuning trained the classifier, final two encoder blocks and final normalisation. SigLIP's pretrained pooling head stayed frozen, but gradients passed through it. Both backbones stayed in evaluation mode, with gradients enabled for the intended parameters. Classifier initialisation was paired within each backbone; different feature dimensions prevented identical heads across models.

Training minimised cross-entropy using AdamW: classifier learning rate 0.001, backbone rate 0.00001, weight decay 0.01, batch size 8, gradient clipping at norm 1 and CUDA FP16. Learning rates were fixed. Training allowed 10 epochs, with patience 3 for strict validation macro-F1 improvement. Checkpoint ties used lower validation loss, then the earlier epoch. Three seeds (17, 42, 2026), two settings and two models produced 12 fresh runs. No hyperparameter search was performed; settings were not separately tuned for each model.

Full-epoch pilots passed for both models/settings, alongside 35 implementation, protocol and analysis tests. Execution used an RTX 3050 Laptop GPU (4 GB), PyTorch 2.6.0, CUDA 12.4 and Transformers

4.57.1. Configurations, histories, hashes and predictions support reproduction. Controlled seeds do not guarantee bit-for-bit CUDA reproducibility. Codex assisted implementation, analysis and writing.

A subsequent unchanged notebook completed all 10 code cells and 12 new runs locally without errors; six setup checks also passed. Its separate partial-model mean macro-F1 scores were 97.67% (DINOv2) and 96.90% (SigLIP 2), showing rerun variability. These do not replace the original results below. Hosted Colab remains untested.

5. Evaluation design

Macro-F1 was the main metric because it gives each class equal weight. Accuracy, per-class precision/recall/F1, confusion matrices and examples were also used. Predicting the most frequent training class provided the majority baseline. All 12 runs finished before selected checkpoints were locked and new test inference began; earlier test exposure still applies.

Means and sample standard deviations across three seeds describe training variability. The reference comparison was partial SigLIP 2 minus partial DINOv2 at seed 17. A paired, class-stratified bootstrap used 2,000 resamples, seed 20261023 and shared image indices to calculate a descriptive 95% percentile interval. This interval concerns that pair, not the three-seed mean. Unknown plant/farm dependence was not corrected.

Raw training totals/means and peak allocated CUDA memory measured cost. Frozen SigLIP's seed-17 epoch 9 took 809.10 seconds, while its other epochs took about 13 seconds; the cause is unknown. Before new test evaluation, a documented descriptive amendment added observed epoch medians, quartiles and ranges. Figure 1 shows pooled epoch medians and interquartile ranges. All raw durations were retained without exclusions or corrections. These are observations, not controlled hardware benchmarks.

Selected checkpoints were scored on clean training/validation images, reusing saved test scores. Training loss with augmentation cannot be directly compared with clean validation loss. Reference-seed blur (radius 1) and brightness ×0.7 tested synthetic sensitivity rather than field robustness.

6. Results and interpretation

Both partial models averaged 96.90% macro-F1, differing by less than 0.01 percentage points (Table 3). SigLIP's frozen features exceeded frozen DINOv2 by 6.18 points. Partial fine-tuning improved DINOv2 by 6.18 points but gave no meaningful average gain for SigLIP.

Configuration	Accuracy (%)	Macro-F1 (%)	Mean training (s)
DINOv2 frozen	90.62 ± 2.07	90.71 ± 2.05	102.8
DINOv2 partial	96.88 ± 1.56	96.90 ± 1.55	96.2
SigLIP 2 frozen	96.88 ± 1.35	96.90 ± 1.34	387.5
SigLIP 2 partial	96.88 ± 0.78	96.90 ± 0.78	195.9

Table 3. Means ± sample SD across three seeds. Training times exclude validation/loading/checkpoint overhead and retain long observations. The majority baseline scored 33.59% accuracy and 16.76% macro-F1.

Partial SigLIP-minus-DINO differences were -0.78, +1.55 and -0.77 points for seeds 17, 42 and 2026 respectively. The seed-17 bootstrap interval was [-3.13, +1.57] points, including zero. Neither the seed pattern nor this interval establishes superiority.

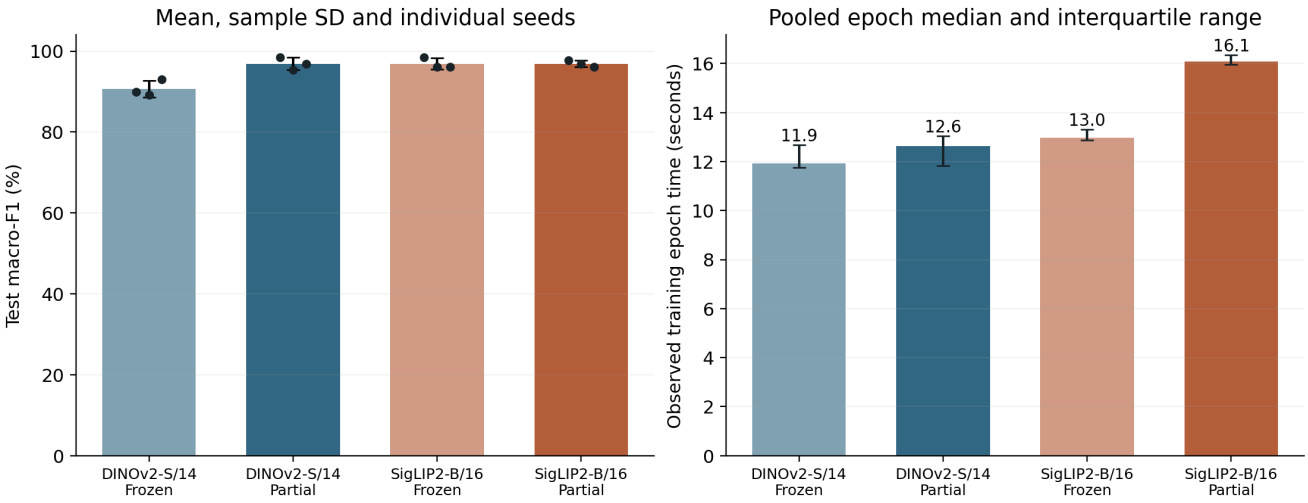


Figure 1. Mean test macro-F1, seed standard deviations and individual runs; median observed training epoch time with interquartile range. Raw totals and ranges accompany the results.

Partial DINOv2/SigLIP median epoch times were 12.62/16.08 seconds, with peak allocated memory 288/722 MB. SigLIP's larger model did not improve mean accuracy. Long epochs included 809.10 seconds for frozen SigLIP and 184.36 seconds for partial SigLIP; causes remain unknown. Raw totals should not be treated as controlled speed comparisons.

Clean train/validation/test macro-F1 averaged 98.17/98.51/96.90% for partial DINOv2 and 98.91/98.26/96.90% for partial SigLIP. The train-test gaps were 1.27 and 2.01 points. Later epochs deteriorated: SigLIP seed 17 validation loss rose from 0.062 at selected epoch 6 to 0.375 at epoch 9 while training loss fell. This indicates possible overfitting or instability. Validation selection retained the earlier checkpoint. Small selected-checkpoint gaps cannot rule out overfitting; supplementary curves show every run.

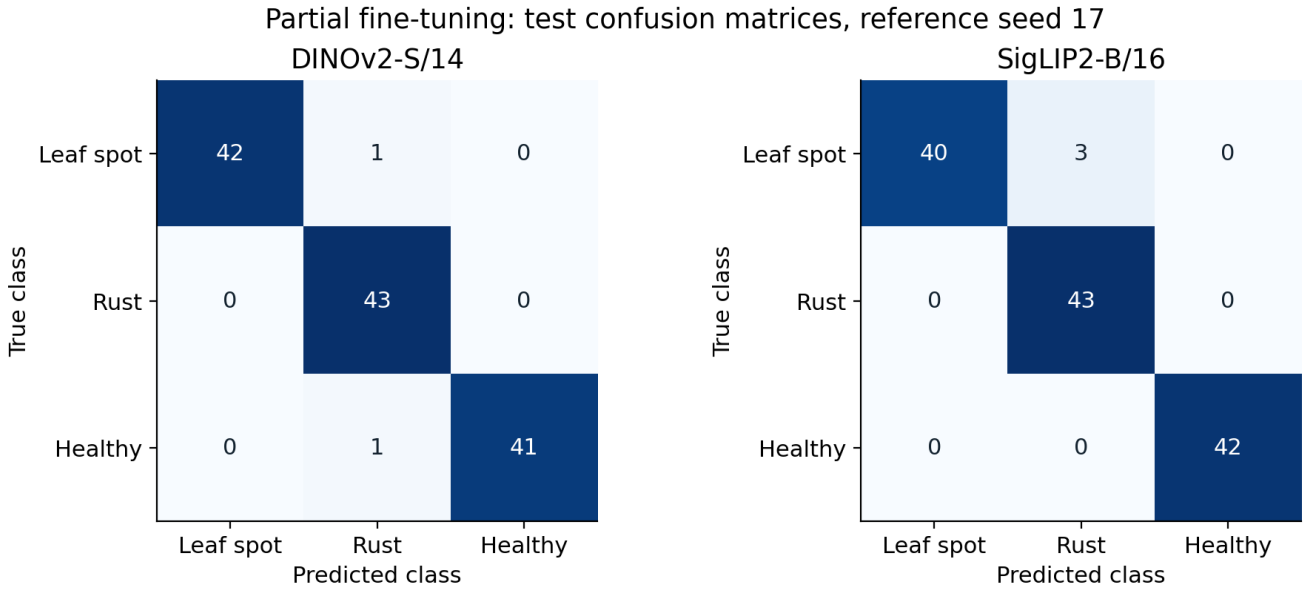


Figure 2. Test confusion matrices for the partial models at reference seed 17. Values are image counts.

Reference SigLIP misclassified three leaf-spot images as rust. Class F1 scores were 96.39%, 96.63% and 100% for leaf spot, rust and healthy respectively. Rust precision was 93.48%, recall 100%; leaf-spot recall was 93.02%. DINOv2 made two errors, with class F1 scores 98.82%, 97.73% and 98.80% in the same order. Neither labelled a diseased leaf healthy (0/86). Healthy-as-diseased counts were 0/42 for SigLIP and 1/42 for DINOv2. One SigLIP error had 98.65% confidence. All per-class metrics accompany the code.

Under blur/darkening, reference SigLIP macro-F1 was 96.07%/97.67%, versus 97.67% clean. DINOv2 scored 97.67%/98.45%, versus 98.45% clean. These checks cannot establish field reliability. With 128 test images, each error changes accuracy by 0.78 points. Three seeds and no external validation limit conclusions.

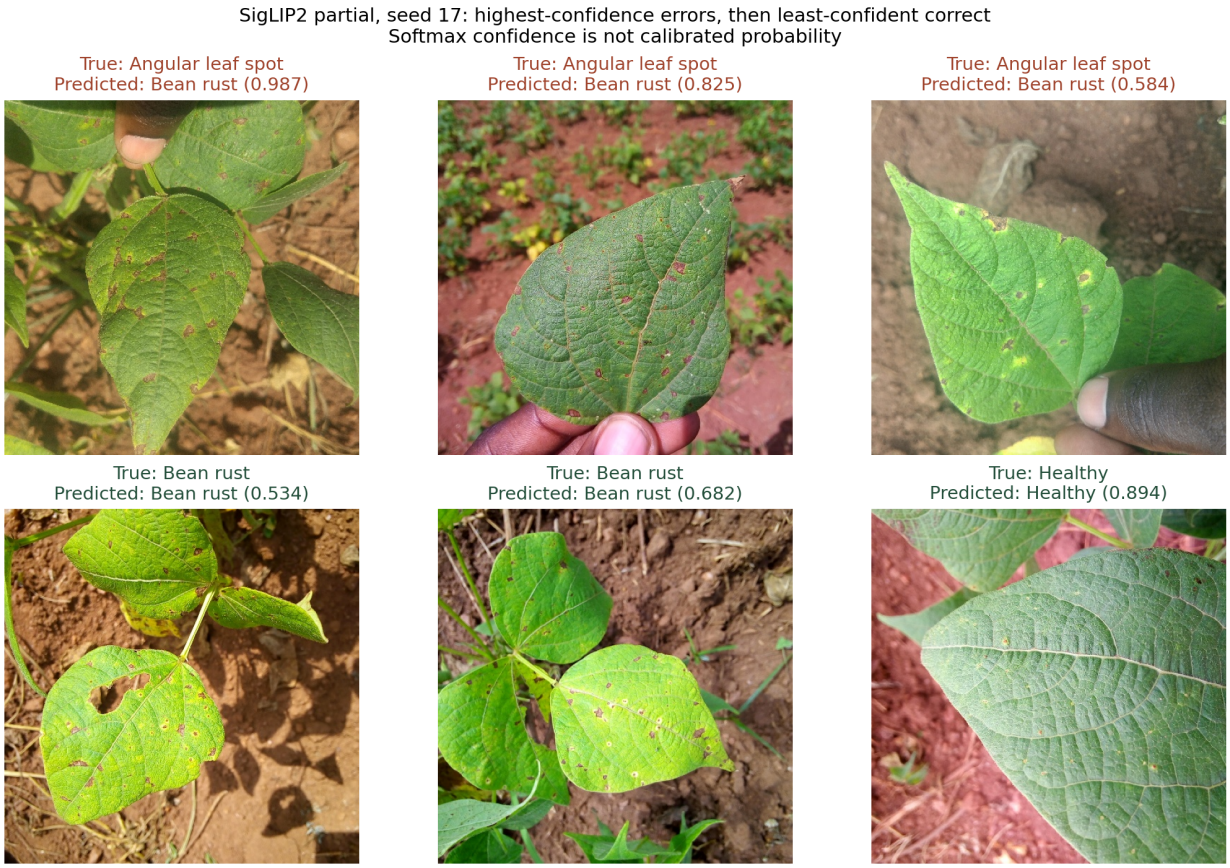


Figure 3. SigLIP 2 partial predictions: highest-confidence errors, then lowest-confidence correct cases to make six examples. Scores are uncalibrated.

7. Ethical and societal considerations

The model could help people recognise visible symptoms where agricultural advice is difficult to access. Errors have different consequences: missing disease could delay expert assessment, while false alarms could cause expense or incorrect treatment. The observed error counts do not measure harm in practice. Zero disease-as-healthy errors does not establish zero risk. Any future system should support expert judgement and avoid automatically prescribing treatment.

The dataset covers one crop and only two diseases. Other diseases, pests or nutrient problems would still be forced into one of the three labels. Confidence does not establish correctness, especially on unfamiliar images. Before deployment, testing should cover independent farms, seasons, devices and disease presentations. A rejection or expert-referral policy would need separate validation and calibration.

Balanced classes do not show fairness across farmers. Missing farm, location and demographic information prevents subgroup evaluation. Pretraining data may also represent some regions or socioeconomic groups better than others. SigLIP 2's reported fairness improvements on other benchmarks [1] do not establish fairness here. Local testing and agricultural practitioners' input are needed. Phone quality, connectivity and access to advice may affect who benefits.

Makerere AI Lab, annotators and model authors need attribution. Dataset MIT and model Apache-2.0 notices accompany the project. Public availability does not establish consent for unrelated future collection or use; new data should minimise personal or farm-identifying information. Reusing pretrained models and

limiting adaptation reduced the training performed here, but local runtime excludes pretraining. Energy use and carbon emissions were not measured. Saved code, predictions and the assistance disclosure document how the work was produced.

8. Conclusion

SigLIP 2 was accessible and feasible, and provided better frozen features. The partial-model means were effectively tied. Partial DINOv2 offers comparable accuracy with a smaller model and lower memory requirements in this exploratory, resource-limited setting. Next steps are to test images from independent farms and tune hyperparameters using validation data.

References

[1] Tschannen, M., et al. (2025). SigLIP 2: Multilingual Vision-Language Encoders with Improved Semantic Understanding, Localization, and Dense Features. Paper.

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[3] Makerere AI Lab (2020). Bean disease dataset. Repository.

[4] Makerere AI Lab (n.d.). Beans dataset card. Accessed 24 October 2026. Dataset.

[5] Google (2025). SigLIP 2 B/16-224 model and processor. Model card. Processor.

[6] Hugging Face (2025). Transformers 4.57.1 SigLIP implementation. Source.

[7] Meta AI (n.d.). DINOv2-Small configuration and processor. Configuration. Processor.